Lecture 10- filled in

April 10, 2026 8:29 AM

Cor.test (x, y)
cor.test (y, x)

Ho: Rho = 0
HA: Rho ≠ 0

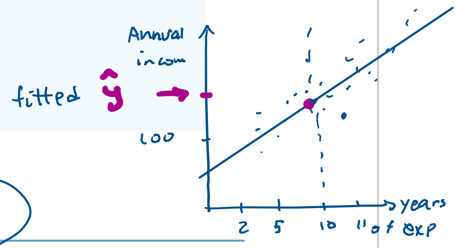
Lecture 10: Linear Regression

Building predictive models from data

x = independent : years of exp...
y = dependent : Annual income.

Learning Objectives

- Distinguish between correlation and regression and explain when each is appropriate
- Interpret the intercept and slope of a fitted regression line in context
- Understand the Ordinary Least Squares (OLS) criterion and how the line of best fit is derived
- Check model assumptions using residual diagnostics
- Evaluate model fit using R^2 and RMSE
- Fit, summarise, and visualise a linear regression model in R



1 From Correlation to Regression

$y = mx + b$ (5, 10)

Last lecture we used Pearson's *r* to measure how strongly two variables move together. Correlation answers: *is there a relationship, and how strong is it?* Regression goes further, it asks: *given a value of x, what value of y should we predict?*

$SSR = \Sigma(\underbrace{y - \hat{y}}_{e_i})^2$ = Minimized

e_i = error
residual

SSR

Correlation	Regression
Measures strength & direction of a linear relationship	Models the relationship as an equation
Result: a single number, r ∈ [–1, 1]	Result: a fitted line with slope and intercept
Symmetric, no distinction between x and y	Asymmetric, x predicts y
Does not produce predictions	Generates predictions for new data

Key insight

The coefficient of determination R^2 in regression equals r^2 from correlation.

```
Multiple R-squared:  0.7528,  Adjusted R-squared:  0.7446
```

one x

more than one x $(x_1, x_2, \dots)$

Interpret: About 75% of the variation in mpg is explained by the variation in car wt (on average)

```
F-statistic: 91.38 on 1 and 30 DF,  p-value: 1.294e-10
```

$\begin{cases} H_0: \beta_1 = \beta_2 = \beta_3 = \dots = 0 \\ H_A: \text{At least one slope} \neq 0 \end{cases}$

1

2 The Simple Linear Regression Model

2.1 The Equation

A simple linear regression model with one predictor has the form:

$$\boxed{Y = \beta_0 + \beta_1 X + \varepsilon}$$

for population

β_0 : intercept of "the regression" line

β_1 : slope of "the regression" line

where:

- **Y** is the response (dependent) variable — the thing we want to predict
- **X** is the predictor (independent / explanatory) variable
- $\boldsymbol{\beta_0}$ (beta-zero) is the intercept — the predicted value of Y when X = 0
- $\boldsymbol{\beta_1}$ (beta-one) is the slope — the change in Y for each one-unit increase in X
- $\boldsymbol{\varepsilon}$ (epsilon) is the error term — the part of Y not explained by X, assumed $\varepsilon \sim N(0, \sigma^2)$

Once we fit the model from data we obtain ~~*estimated* coefficients~~ b_0 and b_1, giving the **fitted line:**

$$\hat{y} = b_0 + b_1 x$$

$\hat{y}$ = predicted / fitted y (comes from the equation)

2.2 Residuals

For each observation, the **residual** is the difference between the actual and fitted value:

$$e_i = y_i - \hat{y}_i$$

y_i = actual (observed in sample)

Residuals are the *observed error* in our model. A positive residual means the model under-predicted; a negative residual means it over-predicted. Residuals are central to both fitting the model and checking its assumptions.

3 Ordinary Least Squares (OLS)

3.1 The OLS Criterion

We want the *best* line through the data. But what does best mean? OLS defines it as the line that minimises the **Sum of Squared Residuals (SSR):**

$$SSR = \sum (y_i - \hat{y}_i)^2 = \sum e_i^2$$

$\frac{d}{dx} = 0$

Why squared? Three good reasons:

- Prevents positive and negative errors cancelling out
- Makes the criterion differentiable, we can find the derivative analytically
- Penalises large errors more heavily than small ones

3.2 The OLS Formulas

Taking the derivative of SSR with respect to β_0 and β_1 and setting to zero yields closed-form solutions:

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{Sxy}{Sxx}$$

Residual = error
OLS = method to minimize errors
Best line = smallest total squared error

$$b_0 = \bar{y} - b_1 \bar{x}$$

Notice that the line of best fit always passes through the point $\boxed{(\bar{x}, \bar{y})}$ the means of x and y. R computes all of this for you, but understanding the formula helps you reason about what the coefficients actually represent.

4 Fitting a Model in R

4.1 The lm() Function

Linear models in R are fitted with lm(). The first argument uses R's formula syntax: response ~ predictor.

```
# Fit a simple linear regression model
model <- lm(mpg ~ wt, data = mtcars)

# Print estimated coefficients
coef(model)

# Full statistical summary
summary(model)
```

```
coef(model)
(Intercept)          wt
  37.285126   -5.344472
```

$\hat{y} = 37.29 - 5.344x$ line of best fit

3.920

$x = 3920$ lb

$\widehat{mpg} = 16.3$ Mpg

4.2 Reading the summary() Output

The summary() output contains several key sections:

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  37.2851     1.8776  19.858  < 2e-16
wt           -5.3445     0.5591  -9.559 1.29e-10
```

$H_0: \beta_1 = 0$

$H_A: \beta_1 \neq 0$

reject Ho

There is a relationship

Section	What it tells you
Coefficients table	Estimated β_0 and β_1, their standard errors, t-statistics, and p-values
Residuals section	Five-number summary of the residuals — check for rough symmetry around 0 Normality Mean ≃ Ø
Multiple R^2 ← one var	Proportion of variance in Y explained by the model (0 to 1)
Adjusted R^2 ← Multiple var	R^2 penalised for number of predictors — more useful when comparing models
F-statistic ← Multiple var	Tests whether the model as a whole is significantly better than the null (no predictors)
Residual Std. Error RSE : the smaller the better	Estimate of σ, the spread of residuals around the regression line (Average size of prediction error)

```
Residual standard error:
3.046
```

(on average how much error to expect when predicting with this model)

4.3 Plotting the Regression Line

```
# Base R scatter plot with regression line
plot(mpg ~ wt, data = mtcars,
```

```
    main = "Car Weight vs. Fuel Efficiency",
    xlab = "Weight (1000 lbs)", ylab = "Miles per Gallon",
    pch = 16, col = "steelblue")
abline(model, col = "firebrick", lwd = 2)

# ggplot2 equivalent
library(ggplot2)
ggplot(mtcars, aes(x = wt, y = mpg)) +
  geom_point(colour = "steelblue", size = 2) +
  geom_smooth(method = "lm", se = TRUE, colour = "firebrick") +
  labs(title = "Car Weight vs. Fuel Efficiency",
     x = "Weight (1000 lbs)", y = "Miles per Gallon")
```

5 Model Evaluation

5.1 R^2 — Coefficient of Determination

R^2 measures the proportion of total variability in Y that is explained by the regression model:

$$R^2 = 1 - SSR / SST$$

where $SST = \sum(y_i - \bar{y})^2$ is the total sum of squares. $R^2 = 0$ means the model explains nothing; $R^2 = 1$ means a perfect fit.

> **Important caveat**
> A high R^2 does not mean the model is correct — it just means the predictors explain a lot of the variance. Always check your residual plots. A model can have $R^2 = 0.99$ and still violate every assumption.

5.2 Root Mean Squared Error (RMSE)

RMSE is expressed in the same units as Y, making it interpretable as the typical prediction error:

$$RMSE = \sqrt{SSR / n}$$

```
# Extract fitted values and compute RMSE manually
fitted_vals <- fitted(model)
residuals_  <- residuals(model)
rmse        <- sqrt(mean(residuals_^2))
cat("RMSE:", round(rmse, 3), "\n")
```

Fitted: all $\hat{y}$

$\sqrt{\left(\frac{SSR}{n}\right)}$

RMSE is the average prediction error.
RSE is the same idea, but adjusted for how many parameters we estimated

```
RMSE> sqrt(mean((model$residuals)^2))
[1] 2.949163
```

RSE = 3.02

6 Checking Model Assumptions

Linear regression relies on four key assumptions — often remembered by the acronym **LINE**:

Letter	Assumption	What to look for
L	**Linearity**	Residuals vs Fitted plot — no curved pattern the more random, the better
I	**Independence**	Residuals vs order plot — no trend over time or index
N	**Normality of residuals**	Q-Q plot — points follow the diagonal line closely
E	**Equal variance (Homoscedasticity)**	Scale-Location plot — constant spread, no funnel shape

6.1 Diagnostic Plots in R

R generates four standard diagnostic plots from a fitted model with a single call:

```
# Display all four diagnostic plots in one figure
par(mfrow = c(2, 2))
plot(model)
par(mfrow = c(1, 1))  # reset layout
```

Alternatively, you can inspect individual plots:

```
# Residuals vs Fitted  (checks linearity & homoscedasticity)
plot(fitted(model), residuals(model),
     xlab = "Fitted Values", ylab = "Residuals",
     main = "Residuals vs Fitted")
abline(h = 0, lty = 2, col = "grey")

# Normal Q-Q plot  (checks normality of residuals)
qqnorm(residuals(model))
qqline(residuals(model), col = "firebrick")
```

7 Making Predictions

Once you have a fitted model you can predict Y for new X values using predict():

```
# Predict mpg for cars weighing 2.5 and 4.0 (1000 lbs)
new_data <- data.frame(wt = c(2.5, 4.0))
predict(model, newdata = new_data)

# With 95% prediction intervals
```

```
predict(model, newdata = new_data, interval = "prediction", level = 0.95)

# With 95% confidence intervals (for the mean response)
predict(model, newdata = new_data, interval = "confidence", level = 0.95)
```

Extrapolation warning

Never predict far outside the range of your training data. The linear relationship observed in your data may not hold elsewhere, and the model has no information about those regions. Always check the range of X in your data before predicting.

8 Practical Example: Full Workflow in R

Here is a complete worked example using the built-in mtcars dataset, predicting fuel efficiency (mpg) from car weight (wt).

```
# ── 1. Explore ──────────────────────────────────────
head(mtcars[, c("mpg", "wt")])
cor(mtcars$wt, mtcars$mpg)        # Pearson r (from last lecture)

# ── 2. Visualise ────────────────────────────────────
plot(mpg ~ wt, data = mtcars, pch = 16, col = "steelblue",
    main = "Weight vs MPG", xlab = "Weight", ylab = "MPG")

# ── 3. Fit ──────────────────────────────────────────
model <- lm(mpg ~ wt, data = mtcars)
summary(model)

# ── 4. Add fitted line ──────────────────────────────
abline(model, col = "firebrick", lwd = 2)

# ── 5. Diagnose ─────────────────────────────────────
par(mfrow = c(2, 2)); plot(model); par(mfrow = c(1, 1))

# ── 6. Evaluate ─────────────────────────────────────
cat("R-squared:", summary(model)$r.squared, "\n")
cat("RMSE:     ", sqrt(mean(residuals(model)^2)), "\n")

# ── 7. Predict ──────────────────────────────────────
predict(model, newdata = data.frame(wt = 3.5),
        interval = "prediction")
```

Practice Problems:

Thursday online 10:30-12:20

1. Complete the following in R using the built-in airquality dataset. Remove rows with missing values first using air <- na.omit(airquality).
    - Plot Ozone against Temp. Does a linear relationship seem appropriate?
    - Fit a linear model with Ozone as the response and Temp as the predictor. Write out the fitted equation with the estimated coefficients.
    - Interpret the slope coefficient in plain English.
    - Report R^2 and RMSE. How well does the model fit?
    - Produce the four diagnostic plots. Are any assumptions visibly violated?
    - Predict the ozone level when temperature is 90°F. Include a 95% prediction interval.
    - (Extension) What happens if you predict for a temperature of 150°F? Why is this problematic?

2. With the given dataset:

x <- c(10, 20, 30, 40, 50, 60, 70, 80)

y <- c(15, 25, 28, 40, 45, 52, 60, 65)

df <- data.frame(x, y)

Questions

Interpret the slope in context.

What does the intercept represent? Is it meaningful?

How well does the model fit the data?

Predict runtime when input size is 100
Are the model residuals normal?
Report R^2 and RMSE. How well does the model fit?

3. Lines of Code vs Number of Bugs (Outlier Case)

Dataset

loc <- c(100, 200, 300, 400, 500, 600, 700, 2000)

bugs <- c(5, 9, 15, 20, 22, 30, 35, 150)

df3 <- data.frame(loc, bugs)

Investigate the requirements of linear regression, fit the model.

Discussion Questions

a) Identify the outlier.
b) How does the outlier affect the slope?
c) What do residual plots show?

d) Should we remove the outlier? Why or why not?

4. Nonlinear Relationship (Model Misfit)

Dataset

```
x <- c(1,2,3,4,5,6,7,8)
y <- c(2,4,7,11,18,29,45,70)
df4 <- data.frame(x,y)
```

Fit the model

```
model4 <- lm(y ~ x, data = df4)
summary(model4)
plot(model4)
```

Questions

a) Does a linear model fit well?

b) What pattern appears in residuals?

c) What does this suggest about the relationship?

d) What type of model might be more appropriate?

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5. Weak Relationship (Low R-squared)

Dataset

```
x <- c(1,2,3,4,5,6,7,8,9,10)
y <- c(5,7,6,8,7,9,8,10,9,11)
df5 <- data.frame(x,y)
```

Fit the model

Questions

1. What does a low R-squared indicate?
2. Is the model useful for prediction?
3. Can the slope still be statistically significant?

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